

claude-windows / 8d4866bf-8edd-4da9-909e-022dde12675

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022dde12675\subagents\workflows\wf_3d035317-127\agent-a76fce09473a7dbdf.jsonl`  
- SHA-256: `c873b847c3dffb21ec2fb45f330b6a288e3fadf998cfb03469dc9d41b18d05ad`  
- Source modified: `2026-06-22T09:50:20+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_3d035317-127`  
- session_id: `8d4866bf-8edd-4da9-909e-022dde12675`
```

Transcript

1. user (2026-06-22T09:46:23.305Z)

You are fixing staleness in ONE doc of the KhelSutra repo, applying an audit that already ran. Be SURGICAL: minimal, targeted edits that match the audited fix; preserve the doc's voice, structure, and markdown formatting. Do NOT wholesale-rewrite a doc.

KhelSutra repo root: C:\Users\avidu\Projects\kheLSutra
ENGINE repo root (sibling, for cross-repo verification): C:\Users\avidu\Projects\badminton-highlight-indexer (origin/master freshly fetched)
Today: 2026-06-22. KhelSutra is on branch docs/freshness-sweep (off master).

YOUR DOC (edit ONLY this file): C:\Users\avidu\Projects\kheLSutra\docs\ANNOTATE_AND_TRAIN.md

The audit findings are in this JSON file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022dde12675\tasks\w3dbzrpby.output
Shape: { result: { perDoc: [{doc, overallStatus, findings, summary}], staleFindings: [{doc, claim, docLocation, verdict, evidence, suggestedFix, severity}] } }.
Read it and select ONLY entries where doc === "docs/ANNOTATE_AND_TRAIN.md". Your worklist = that doc's staleFindings. Also skim that doc's perDoc.findings for "unverifiable" items ? do NOT act on those unless you can now positively verify them.

For each stale finding:

1. RE-VERIFY it yourself against the LIVE repo before editing ? Read the doc lines + the cited code. For any claim about the ENGINE (paths like backend/main.py, /api/*, native_wasb_runner.py, auth/CORS), verify against the engine repo's origin/master: e.g. ``git -C /c/Users/avidu/Projects/badminton-highlight-indexer grep -n "<symbol>" origin/master -- backend/`` or ``git -C ... show origin/master:backend/main.py``. Trust nothing blindly ? the audit itself may be slightly off and engine line numbers DRIFT between PRs.
2. If confirmed, apply the suggestedFix as a SURGICAL Edit. For brittle ENGINE line-number anchors, prefer SYMBOL-based references (e.g. ``backend/main.py::get_config``) over chasing exact line numbers ? but only after confirming the symbol exists on engine origin/master.
3. DOCS ONLY: if a fix would require a CODE change (e.g. a comment in site/server.js), do NOT edit code ? record it under codeChangesNeeded.

Hard rules:

- Edit ONLY your assigned doc file. Run NO git WRITE commands (no add/commit/checkout/branch/stash) ? read-only git for verification only.
- Keep status markers HONEST: flip Todo->Done only WITH the PR number / file evidence; never invent a completion.
- If a finding is wrong, not reproducible, or too risky to auto-apply, SKIP it and explain under skipped.
- Do not "improve" beyond the findings; no scope creep.

Return the structured report.

2. assistant (2026-06-22T09:46:26.898Z)

I'll start by reading the audit findings and my assigned doc.

3. user (2026-06-22T09:46:29.203Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-591 of 1722 total (61859 tokens, cap 25000). Call Read with offset=592 limit=591 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```
1      {
2      "summary": "Audit every KhelSutra doc for staleness against the live repo, returning
structured findings",
3      "agentCount": 15,
4      "logs": [
5      "audited 15/15 docs; 44 stale findings; archive-candidates: none"
6      ],
7      "result": {
8      "perDoc": [
9      {
10     "doc": "NEXT_STEPS.md",
11     "overallStatus": "minor-staleness",
12     "summary": "NEXT_STEPS.md is overwhelmingly accurate against master: routes,
file paths, the Worker/D1 names, the 6-language localization, the 3-step wizard, the per-rally
PR numbers (#11/#14/#21/#28/#30/#31), the appliance/per-rally doc statuses, and every still-
pending \"Queued\" item all check out against code and git. The only real staleness is the \"PRs
through #93 merged\" count, which lags by one because PR #94 (the very commit that wrote
\"through #93\") merged afterward; plus a low-severity GCS-artifact filename mismatch. Not an
archive candidate ? it is the live next-steps tracker.",
13     "findings": [
14     {
15     "claim": "PRs through #93 merged (master).",
16     "docLocation": "NEXT_STEPS.md:13",
17     "verdict": "stale",
18     "evidence": "git log --oneline -1 = '50b2bd7 chore(repo): ... correct stale
NEXT_STEPS claims (#94)'; total 94 commits/PRs on master. Ironically #94 is the commit that
updated this line to '#93', then merged itself, so the doc now lags one PR behind master.",
19     "suggestedFix": "Update to 'PRs through #94 merged (master).'",
20     "severity": "medium"
21     },
22     {
23     "claim": "Where we are (2026-06-20): ... khelsutra.guru is LIVE and fully
shipped (landing + /capture checklist + OG image + segmented Request-Access form).",
24     "docLocation": "NEXT_STEPS.md:5-14",
25     "verdict": "correct",
26     "evidence": "site/server.js:98 maps /capture->capture.html;
site/public/og.png + capture.html tracked; index.html:191-193 textareas + lines 537-581
academy/individual/gpu/storage fields. Header date 2026-06-20 is slightly behind the latest work
(PR #94 merged 2026-06-22) but the substance holds.",
27     "suggestedFix": "Optionally bump the 'Where we are' date to 2026-06-22 to
match the latest merged work.",
28     "severity": "low"
29     },
30     {
31     "claim": "On Cloudflare (Worker hidden-king-khelsutra + D1 khelsutra-
access): the form persists to D1 and emails a summary via an Email Routing send binding.",
32     "docLocation": "NEXT_STEPS.md:7-9",
33     "verdict": "correct",
34     "evidence": "site/wrangler.toml: name = \"hidden-king-khelsutra\";
[[d1_databases]] database_name = \"khelsutra-access\"; [[send_email]] name = \"NOTIFY\".
site/src/index.js:80 handles POST /api/request-access -> insert into D1 binding DB.",
35     "suggestedFix": "",
36     "severity": "low"
37     },
38     ]
39     },
40     }
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38         {
39             "claim": "CI gates every PR ? three jobs: full site npm test, web/ SPA build
+ coverage floor, and a real-browser layout check.",
40             "docLocation": "NEXT_STEPS.md:9-12",
41             "verdict": "correct",
42             "evidence": "site/package.json scripts include test (node --test),
test:layout (layout.browser.mjs); web has a coverage floor (#80). PR #94 message confirms 'three
jobs: site npm test = 86 green, web SPA build+coverage, real-browser layout'. Non-browser site
suite ran green (61 tests in the 4 non-layout files).",
43             "suggestedFix": "",
44             "severity": "low"
45         },
46         {
47             "claim": "The form also runs on a dependency-free Node + SQLite host
(site/server.js), live on the DO droplet -> http://168.144.126.176:8787 (systemd khelsutra-
site).",
48             "docLocation": "NEXT_STEPS.md:11-13",
49             "verdict": "correct",
50             "evidence": "site/server.js exists, uses node:sqlite (header comment line
2). IP 168.144.126.176 + port 8787 + 'khelsutra-site' all appear in
docs/LOCAL_APPLIANCE_ARCHITECTURE.md:48 ('/srv/khelsutra-site, node :8787') and deploy refs.
Live-status is external/unverifiable from the repo, but the host code and identifiers are
correct.",
51             "suggestedFix": "",
52             "severity": "low"
53         },
54         {
55             "claim": "Docs: docs/REQUEST_ACCESS_FORM.md, docs/PORTABILITY.md.",
56             "docLocation": "NEXT_STEPS.md:13-14",
57             "verdict": "correct",
58             "evidence": "git ls-files: both docs/REQUEST_ACCESS_FORM.md and
docs/PORTABILITY.md are tracked.",
59             "suggestedFix": "",
60             "severity": "low"
61         },
62         {
63             "claim": "DONE ? Friendly & Multilingual (ARCHIVED 2026-06-22): localized to
all 6 languages (en/kn/hi/es/da/id); full record docs/archives/FRIENDLY_AND_MULTILINGUAL.md.",
64             "docLocation": "NEXT_STEPS.md:17-21",
65             "verdict": "correct",
66             "evidence": "docs/archives/FRIENDLY_AND_MULTILINGUAL.md is tracked; git log
shows PR #93 'docs: archive the Friendly & Multilingual tracker'. web/src/i18n/index.ts:25
SUPPORTED_LOCALES = [en,kn,hi,es,da,id]; PR #87 added es/da/id to the marketing site.",
67             "suggestedFix": "",
68             "severity": "low"
69         },
70         {
71             "claim": "Interactive per-rally selection MVP/M0 COMPLETE; Done: P0 design
(#11), P1 scaffold (#14), P2 grid+hook (#21), P3 parseQuery query bar (#28), P4 SelectionFooter
(#30), P5 real load+download (#31). Tracked doc docs/PER_RALLY_SELECTION.md (IN PROGRESS).",
72             "docLocation": "NEXT_STEPS.md:32-41",
73             "verdict": "correct",
74             "evidence": "git log: cb74ce5 #11 'tracked design ... RallyPicker'; aac7e0c
#14 'P1 scaffold'; 62598e7 #21 'P2 selectable rally grid + hook'; d537d19 #28 'P3 deterministic
parseQuery() query bar'; aff8de8 #30 'P4 SelectionFooter + min_shot_count warning'; 8bf200d #31
'P5 load real rallies + Build->compile->preview->download'. docs/PER_RALLY_SELECTION.md status
header = 'IN PROGRESS ? MVP/M0 (P1-P5) COMPLETE'.",
75             "suggestedFix": "",
76             "severity": "low"
77         },
78         {
79             "claim": "P2c first-run setup wizard ? 3-step SetupWizard from
/api/capabilities, 6 languages.",
80             "docLocation": "NEXT_STEPS.md (implied via track) and verified for the
merged #88",

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81         "verdict": "correct",
82         "evidence": "web/src/components/SetupWizard.tsx:22 'const STEPS = 3'; line 1
comment 'Three quick steps'; line 3 'Localized (en/kn/hi/es/da/id)'; web/src/api/client.ts:139
GET /capabilities. git: 7ae9cee #88.",
83         "suggestedFix": "",
84         "severity": "low"
85     },
86     {
87         "claim": "Single-box appliance ? Tracked design
docs/LOCAL_APPLIANCE_ARCHITECTURE.md (DRAFT, owner-gated).",
88         "docLocation": "NEXT_STEPS.md:52-58",
89         "verdict": "correct",
90         "evidence": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md status header: 'Status:
DRAFT ? owner-gated (auth choice, retention, cost ceiling, GPU host, P5 bridge)'.",
91         "suggestedFix": "",
92         "severity": "low"
93     },
94     {
95         "claim": "Queued: Always-Use-HTTPS + HSTS (Cloudflare) and Cloudflare Web
Analytics token paste into site/public/index.html ? both still pending.",
96         "docLocation": "NEXT_STEPS.md:44-45",
97         "verdict": "correct",
98         "evidence": "No Strict-Transport-Security/HSTS header found in site/*.js or
wrangler.toml. site/public/index.html:629 still has data-cf-beacon token 'REPLACE_WITH_CF_TOKEN'
(commented-out block). Both correctly listed as not-yet-done.",
99         "suggestedFix": "",
100        "severity": "low"
101    },
102    {
103        "claim": "Droplet -> production: add VM-path SMTP so the Node host also
emails (currently noopNotify).",
104        "docLocation": "NEXT_STEPS.md:46-47",
105        "verdict": "correct",
106        "evidence": "site/server.js:14 imports noopNotify; line 24 'const notify =
noopNotify(); // VM host: no email yet (add SMTP later)'.",
107        "suggestedFix": "",
108        "severity": "low"
109    },
110    {
111        "claim": "Org consolidation DONE; khelsutra (this site repo) transfer is an
owner step still PENDING (unchecked).",
112        "docLocation": "NEXT_STEPS.md:65-73",
113        "verdict": "correct",
114        "evidence": "git remote -v shows origin =
https://github.com/avidullu/khelsutra.git (NOT under Khelsutra org) ? consistent with the '?
owner step pending' marker. Other repos already moved: docs/DRY_RUN_CUJ.md:49 links
github.com/Khelsutra/badminton-highlight-indexer#289.",
115        "suggestedFix": "",
116        "severity": "low"
117    },
118    {
119        "claim": "A real-UI CUJ screencast exists -> promo_cuj.mp4/.gif in
gs://khelsutra-rally-corpus/promo/.",
120        "docLocation": "NEXT_STEPS.md:29-31",
121        "verdict": "unverifiable",
122        "evidence": "No GCS access to confirm the object. The companion doc
docs/DRY_RUN_CUJ.md:85 names the actual artifacts 'khelsutra-bucket-cuj-screencast.mp4' and ':60
khelsutra-i18n-screencast.mp4' in the same gs://khelsutra-rally-corpus/promo/ prefix;
'promo_cuj.mp4' appears nowhere else in the repo, suggesting a possible filename mismatch.",
123        "suggestedFix": "Reconcile the filename with docs/DRY_RUN_CUJ.md (likely
'khelsutra-bucket-cuj-screencast.mp4') or confirm 'promo_cuj.mp4' is a real object in the
bucket.",
124        "severity": "low"
125    },
126    {

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127         "claim": "Control plane (control-plane/) at beta.",
128         "docLocation": "NEXT_STEPS.md:62",
129         "verdict": "correct",
130         "evidence": "control-plane/ directory does not exist in the repo (test -d
looking/aspirational item, not a claim of present existence ? so it is not stale as written.",
131         "suggestedFix": "",
132         "severity": "low"
133     }
134 ]
135 },
136 {
137     "doc": "README.md",
138     "overallStatus": "significant-staleness",
139     "summary": "The repo-structure list, deploy references, and canonical-doc
pointers are all correct, but the README's headline status is badly out of date: it still calls
the repo a \"pre-alpha scaffold\" whose SPA has yet to land \"per milestone M1\", whereas the
web/ SPA is in fact built, multilingual, and shipped (per-rally MVP P0?P5, P2c wizard, in-UI
upload, 6 languages; khelsutra.guru is LIVE). The control-plane/ directory it lists also does
not yet exist.",
140     "findings": [
141     {
142         "claim": "Status: pre-alpha scaffold. SPA lands in `web/` per execution-plan
milestone M1.",
143         "docLocation": "README.md:9",
144         "verdict": "stale",
145         "evidence": "web/src/ has 51 tracked source files incl. web/src/App.tsx;
`git log --oneline -- web/` = 31 commits; merged PRs show the SPA is feature-complete: P2c
first-run wizard (#88), in-UI file upload T4 (#66), per-rally selection MVP P0?P5
(#11/#14/#21/#28/#30/#31), 6-language catalogs (#67). NEXT_STEPS.md:7 states 'khelsutra.guru is
LIVE and fully shipped'. UI_ALPHA_EXECUTION_PLAN.md:36 defines M1 only as the initial
'Vite+React+TS+Tailwind scaffold', which is long past.",
146         "suggestedFix": "Replace with a current status, e.g.: 'Status: alpha ? the
web/ SPA is built and multilingual (6 languages), with the per-rally selection MVP and first-run
setup wizard shipped; khelsutra.guru / app.khelsutra.guru are live (Cloudflare Pages +
cloudflared tunnel + Access). See NEXT_STEPS.md.'",
147         "severity": "high"
148     },
149     {
150         "claim": "the web app (alpha SPA) now, the hosted control plane at beta",
151         "docLocation": "README.md:4",
152         "verdict": "stale",
153         "evidence": "This tagline pairs with line 9's 'pre-alpha scaffold', which is
contradicted by the shipped, live SPA (see web/ evidence above and NEXT_STEPS.md:7). The 'alpha
SPA now' framing is directionally right but undercut by the stale 'pre-alpha scaffold' status it
sits beside.",
154         "suggestedFix": "Keep 'web app (alpha SPA) now' but align the rest of the
intro/status so it no longer describes the SPA as not-yet-landed.",
155         "severity": "medium"
156     },
157     {
158         "claim": "`control-plane/` ? (beta) Cloud Run FastAPI control plane",
159         "docLocation": "README.md:17",
160         "verdict": "stale",
161         "evidence": "No control-plane/ directory exists: `ls control-plane` -> 'No
such file or directory'; `git ls-files control-plane/` returns nothing. Top-level tracked dirs
are deploy/, docs/, locales/, site/, web/ only.",
162         "suggestedFix": "Mark it as a future/planned directory that does not yet
exist (e.g. move it out of the 'Structure' list into a 'Planned (beta)' note), so a reader
doesn't expect the path to be present.",
163         "severity": "medium"
164     },
165     {
166         "claim": "Engine ... sibling repo `badminton-highlight-indexer` (rename
pending).",

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167         "docLocation": "README.md:6",
168         "verdict": "correct",
169         "evidence": "Sibling repo still exists under that name: `ls
C:/Users/avidu/Projects/` shows 'badminton-highlight-indexer'; no rename has occurred, so
'rename pending' holds.",
170         "suggestedFix": "",
171         "severity": "low"
172     },
173     {
174         "claim": "Canonical planning docs live in the engine repo's docs/ for now:
UI_PROPOSAL.md, HOSTING_PLAN.md, UI_ALPHA_EXECUTION_PLAN.md.",
175         "docLocation": "README.md:7-8",
176         "verdict": "correct",
177         "evidence": "All three exist in the engine repo (badminton-highlight-
indexer/docs/UI_PROPOSAL.md, HOSTING_PLAN.md, UI_ALPHA_EXECUTION_PLAN.md present) and are NOT
tracked in khelsutra/docs (`git ls-files docs/` in khelsutra returns none of them; `ls
docs/UI_PROPOSAL.md` -> not found).",
178         "suggestedFix": "",
179         "severity": "low"
180     },
181     {
182         "claim": "`deploy/` ... alpha deploy kit ... DEPLOY_ALPHA.md ... +
cloudflared.config.example.yml",
183         "docLocation": "README.md:15",
184         "verdict": "correct",
185         "evidence": "git ls-files deploy/ shows deploy/DEPLOY_ALPHA.md and
deploy/cloudflared.config.example.yml (plus deploy/droplet/*). DEPLOY_ALPHA.md:1-5 matches the
described 'shareable page, no SSH; SPA on Pages + engine on api.* tunnel + CF Access' content.",
186         "suggestedFix": "",
187         "severity": "low"
188     },
189     {
190         "claim": "`site/` ? public marketing landing page (static; served at
khelsutra.guru)",
191         "docLocation": "README.md:13",
192         "verdict": "correct",
193         "evidence": "site/ exists with site/public/index.html, server.js, etc.;
DEPLOY_ALPHA.md and NEXT_STEPS.md confirm the marketing site is served at khelsutra.guru (apex)
while the SPA is at app.khelsutra.guru.",
194         "suggestedFix": "",
195         "severity": "low"
196     },
197     {
198         "claim": "`web/` ? React SPA (alpha; Vite + React + TS + Tailwind;
Cloudflare Pages build root)",
199         "docLocation": "README.md:14",
200         "verdict": "correct",
201         "evidence": "web/ is a Vite+React+TS+Tailwind SPA (web/README.md:3 'Vite +
React + TS + Tailwind v4'); DEPLOY_ALPHA.md:14 confirms 'app.khelsutra.guru -> Cloudflare Pages
(the web/ SPA)'. The tech-stack/build-root facts are accurate; only the inline 'alpha' maturity
ties into the stale status above.",
202         "suggestedFix": "",
203         "severity": "low"
204     },
205     {
206         "claim": "License: All rights reserved (private; no OSS license).",
207         "docLocation": "README.md:19-21",
208         "verdict": "correct",
209         "evidence": "No LICENSE file is tracked (not present in `git ls-files`);
consistent with a private 'all rights reserved' repo.",
210         "suggestedFix": "",
211         "severity": "low"
212     }
213 ]
214 },

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215     {
216         "doc": "deploy/DEPLOY_ALPHA.md",
217         "overallStatus": "minor-staleness",
218         "summary": "Verified correct except a stale header",
219         "findings": [
220             {
221                 "claim": "header stale deploy live",
222                 "docLocation": "deploy/DEPLOY_ALPHA.md:3",
223                 "verdict": "stale",
224                 "evidence": "droplet/README.md:100 live-verified 2026-06-22",
225                 "suggestedFix": "set header to LIVE",
226                 "severity": "medium"
227             },
228             {
229                 "claim": "host list too narrow",
230                 "docLocation": "deploy/DEPLOY_ALPHA.md:66",
231                 "verdict": "stale",
232                 "evidence": "engine.env.example adds loopback",
233                 "suggestedFix": "add localhost",
234                 "severity": "low"
235             }
236         ]
237     },
238     {
239         "doc": "deploy/droplet/README.md",
240         "overallStatus": "fresh",
241         "summary": "This command-dense droplet quickstart is accurate and matches the
live repo: every cited kit file exists with the described contents, and the engine-behavior
claims (fail-fast boot posture, RALLY_* exposure vars, the Gemini-default-on-CPU gotcha, the
gs:// jail rejection, the --server entryptpoint) all verify against the engine repo and
DEPLOY_ALPHA.md. The only items not locally reproducible are the 2026-06-22 droplet live-run
figures (e.g. \"1313 passed\"), which are an off-box assertion ? they are internally plausible
(engine has ~1269 test functions before parametrization) but cannot be confirmed from these
repos.",
242         "findings": [
243             {
244                 "claim": "Four kit files exist: config.json.example, engine.env.example,
khelsutra-engine.service (and ../cloudflared.config.example.yml), each going to the documented
destination with the documented purpose.",
245                 "docLocation": "deploy/droplet/README.md:9-14",
246                 "verdict": "correct",
247                 "evidence": "git ls-files deploy/ lists all four:
deploy/droplet/{config.json.example,engine.env.example,khelsutra-engine.service} and
deploy/cloudflared.config.example.yml; contents read and match (config.json.example has the
security block + cf_team/aud; engine.env.example has the two RALLY_* exposure vars +
RALLY_APP_HOME; the service binds 127.0.0.1:8000; cloudflared.config routes api. ->
127.0.0.1:8000).",
248                 "suggestedFix": "",
249                 "severity": "low"
250             },
251             {
252                 "claim": "The boot fails fast (exit 1, '[STARTUP] refusing to start') if
edge_auth_enabled=true but the jail or cf_team_domain/cf_access_aud is unset.",
253                 "docLocation": "deploy/droplet/README.md:53-54",
254                 "verdict": "correct",
255                 "evidence": "backend/main.py:857 logs '[STARTUP] refusing to start';
backend/config/startup.py:109 EXPOSED_NO_VIDEO_ROOT and :123 EXPOSED_EDGE_AUTH_INCOMPLETE are
both LEVEL_ERROR (startup.py:41,272-276), while EXPOSED_AUTH_EXTRA_MISSING (PyJWT) is LEVEL_WARN
(:135). From engine PR #294 (commit 0c0e064).",
256                 "suggestedFix": "",
257                 "severity": "low"
258             },
259             {
260                 "claim": "indexing.default_segmenter MUST be gemini on a CPU box; the
shipped default yolo_hybrid needs torch/GPU and every job fails 'No module named torch' without

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it.",
261         "docLocation": "deploy/droplet/README.md:109-113",
262         "verdict": "correct",
263         "evidence": "config.json.example:9-11 sets default_segmenter=gemini. Engine
AppConfig default is yolo_hybrid (backend/config/models.py:128), so an all-defaults load on the
droplet (which hand-writes config.json rather than running the --app LIGHT seeder) does land on
yolo_hybrid. The LIGHT 'motion' default (loader.py:227) only applies via
write_light_default_config (indexer --app first-run, PR #310), not this systemd path ? so the
gotcha is accurate for this deploy.",
264         "suggestedFix": "",
265         "severity": "low"
266     },
267     {
268         "claim": "RALLY_ALLOWED_HOSTS is the load-bearing fix; with edge_auth on,
the Host-header guard defaults loopback-only and 400s every tunneled request until set.",
269         "docLocation": "deploy/droplet/README.md:92, engine.env.example:14-17",
270         "verdict": "correct",
271         "evidence": "backend/config/startup.py:143-147 raises
EXPOSED_HOST_ALLOWLIST_LOOPBACK (LEVEL_WARN) when edge_auth is ON and RALLY_ALLOWED_HOSTS is
unset, with the exact 'will 400 forwarded requests' rationale; backend/main.py reads
RALLY_ALLOWED_HOSTS / RALLY_BIND_HOST / RALLY_CORS_ALLOW_ORIGINS. Matches DEPLOY_ALPHA.md §2.2
repro (400->200).",
272         "suggestedFix": "",
273         "severity": "low"
274     },
275     {
276         "claim": "Pasting a gs:// URI is rejected by the jail on the exposed box;
upload is the alpha path.",
277         "docLocation": "deploy/droplet/README.md:96-98",
278         "verdict": "correct",
279         "evidence": "tests/test_api_endpoints.py:608
test_process_hosted_mode_rejects_nonlocal_uri pins the gs:// -> 400 behavior under
allowed_video_root (referenced at :541 as the counterweight); the jail is required to expose
(config.json.example sets allowed_video_root).",
280         "suggestedFix": "",
281         "severity": "low"
282     },
283     {
284         "claim": "Pages build: Root dir web, Build npm run build, Output web/dist,
with VITE_API_BASE = https://api.khelsutra.guru/api.",
285         "docLocation": "deploy/droplet/README.md:73-76",
286         "verdict": "correct",
287         "evidence": "khelsutra web/ SPA exists (git ls-files web/...);
web/package.json build script = 'vite build'; web/src/api/client.ts:25 reads
import.meta.env.VITE_API_BASE (default '/api'), and client.test.ts:32 exercises the exact split
value https://api.khelsutra.guru/api. Vite default outDir 'dist' relative to root web/ =
web/dist.",
288         "suggestedFix": "",
289         "severity": "low"
290     },
291     {
292         "claim": "The shipped engine is run via the auth extra and python -m
backend.main --server, binding 127.0.0.1:8000.",
293         "docLocation": "deploy/droplet/README.md:33,47-49; khelsutra-
engine.service:26",
294         "verdict": "correct",
295         "evidence": "backend/main.py:867 defines the --server flag and :1107 calls
uvicorn.run(app, host=bind_host, port=port); --setup flag at :865; the service ExecStart uses
python -m backend.main --server.",
296         "suggestedFix": "",
297         "severity": "low"
298     },
299     {
300         "claim": "Live-verified on the droplet (2026-06-22): full suite 1313 passed
+ ruff/mypy clean on the box; boot-posture, Host-guard 400->200, path-jail, T4 upload->jail,

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edge-auth 401-gate exercised live; full DNS->Cloudflare->Access->tunnel->engine chain
confirmed.",
301         "docLocation": "deploy/droplet/README.md:100-106",
302         "verdict": "unverifiable",
303         "evidence": "This is an off-box live-run assertion that cannot be reproduced
from the khelsutra or engine repos here. Sanity check: the engine tests/ dir has ~1269 'def
test_' functions (parametrize expansions push the collected/passed count higher), so '1313
passed' is plausible and consistent, not contradicted. Marked unverifiable per the file:line-
evidence discipline; no sign of staleness.",
304         "suggestedFix": "",
305         "severity": "low"
306     },
307     {
308         "claim": "Cost: Cloudflare Pages + Tunnel + Access are free (Access <= 50
users); standing cost ~ $0 + Gemini per run.",
309         "docLocation": "deploy/droplet/README.md:16-18",
310         "verdict": "unverifiable",
311         "evidence": "External Cloudflare pricing claim; not checkable against the
repo. Consistent with DEPLOY_ALPHA.md:3 ('~$0/mo standing ... + Gemini per-run'). No internal
contradiction.",
312         "suggestedFix": "",
313         "severity": "low"
314     }
315 ]
316 },
317 {
318     "doc": "docs/ANNOTATE_AND_TRAIN.md",
319     "overallStatus": "significant-staleness",
320     "summary": "A DRAFT tracker whose marketing-page deliverables (P0-P4 plus the
real annotator screenshots P5 mentions) are all shipped and merged (#15/#22/#23/#25), yet the
doc was last touched 2026-06-20 (#23) and never updated to reflect that. The core
engine/annotator technical claims (schema bridge, ship-gate refusing a commercial ship, MIT
annotator v1.6.4, guardrails) all still verify as correct ? only their file:line citations have
drifted. Main staleness: status still DRAFT despite shipped work, P5 marked Todo though
annotator screenshots are committed, the doc is NOT actually indexed in docs/README.md (claimed
three times), and the page since grew 6-language localization the tracker never anticipated. It
is an archive candidate once honestly closed out.",
321     "findings": [
322     {
323         "claim": "Status: DRAFT ? owner-gated on scope (header, line 3); Last
updated: 2026-06-20.",
324         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:3",
325         "verdict": "stale",
326         "evidence": "git log --follow shows the marketing-page work merged: P0 doc
(#15, 6b62ebf), own-model page P1-P4 (#22, b1110e1), org link (#23, 87520b8). Per $7, P0-P4 are
all marked Done ? yet the header still reads DRAFT. Lifecycle (line 4) is DRAFT->IN
PROGRESS->DONE; with P0-P4 shipped this is at least IN PROGRESS.",
327         "suggestedFix": "Flip Status to IN PROGRESS (only owner-gated P5-P9 remain)
and bump Last updated; or, if the owner considers the marketing surface complete, mark DONE and
archive per the lifecycle note.",
328         "severity": "medium"
329     },
330     {
331         "claim": "P5 'Capture + commit real screenshots ... replace placeholders' ?
Status ? (Todo), Gated Owner.",
332         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:76",
333         "verdict": "stale",
334         "evidence": "PR #25 (1906c82) 'show the real annotator screenshots'
committed site/public/img/annotate-enable.png (133712 bytes) and annotate-tool.png (153133
bytes); own-model.html:111-112 references them as the real VLC-UI shots (no people/footage). The
annotator-UI half of P5 is done and merged, not Todo.",
335         "suggestedFix": "Mark P5 ? for the annotator-UI screenshots (cite #25); if
footage screenshots with consented people are still pending, split that into its own row so the
done part isn't hidden under a blanket Todo.",
336         "severity": "high"

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337         },
338         {
339             "claim": "Doc is 'index[ed] in docs/README.md' (line 5), §7 P0 'indexed in
docs/README.md' (? , #15), and DoD 'doc indexed in docs/README.md'.",
340             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:5,71,91",
341             "verdict": "stale",
342             "evidence": "grep -ni 'annotate|own.model|train.your|ANNOTATE_AND'
docs/README.md returns NOT FOUND; the docs/README.md 'Tracked work' list (lines 1-30) does not
mention ANNOTATE_AND_TRAIN.md or own-model. The claimed index entry does not exist.",
343             "suggestedFix": "Either add an ANNOTATE_AND_TRAIN.md entry to
docs/README.md's 'Tracked work' list, or correct P0/DoD to reflect that it is not yet indexed.",
344             "severity": "high"
345         },
346         {
347             "claim": "Engine ship-gate (gate_verdict, harness.py:98) currently refuses a
commercial ship regardless of F1: WASB provenance C3 unresolved + F1@tIoU not calibrated
(#172).",
348             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:12,31",
349             "verdict": "correct",
350             "evidence": "training/gen0/harness.py:177 returns {'ship': False, ...}; :175
'WASB checkpoint provenance C3 UNRESOLVED ? commercial ship blocked regardless of F1'; :174
'ship-gate F1@tIoU target NOT YET CALIBRATED ... see issue #172'. Fact correct; the cited line
(:98) has drifted ? gate_verdict is now at :142.",
351             "suggestedFix": "Update the line citations: gate_verdict is at
harness.py:142 (not :98).",
352             "severity": "low"
353         },
354         {
355             "claim": "Default synth path: synth_trajectory_from_labels uses
csv.DictReader reading row['start_time']/row['end_time'] (stub_runner.py:139) ? MATCHES the
annotator header exactly.",
356             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:46",
357             "verdict": "correct",
358             "evidence": "backend/pipeline/detectors/stub_runner.py:124 def
synth_trajectory_from_labels; :137 for row in _csv.DictReader(f); :139
rallies.append((float(row['start_time']), float(row['end_time']))). Annotator header
rally_number,start_time,end_time,... confirmed at rally-annotator vlc/rally_annotator.lua:128
and docs/CSV_FORMAT.md:6. Claim accurate.",
359             "suggestedFix": "",
360             "severity": "low"
361         },
362         {
363             "claim": "Canonical GT reader load_gt_intervals accepts start/end OR
start_time/end_time (gt_loader.py:14-90).",
364             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:47",
365             "verdict": "correct",
366             "evidence": "backend/eval/gt_loader.py:30 _START_NAMES = ('start',
'start_time'); :44 def load_gt_intervals; :70 headerless positional col0,col1 path. Both aliases
accepted as claimed.",
367             "suggestedFix": "",
368             "severity": "low"
369         },
370         {
371             "claim": "Bundle attests no_paid_llm_training_labels:True (harness.py:192)
and label source is human (harness.py:178).",
372             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:33,60",
373             "verdict": "correct",
374             "evidence": "training/gen0/harness.py:236 'no_paid_llm_training_labels':
True; :222 'label_source': 'human (rally-annotator golden CSVs)'. Facts correct; cited lines
(:192, :178) have drifted to :236 and :222.",
375             "suggestedFix": "Update line citations: no_paid_llm_training_labels at
harness.py:236, label_source at :222.",
376             "severity": "low"
377         },
378         {

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379         "claim": "cmd_train (__main__.py:231) pulls labels/, joins video<->label by
file MD5 + stem-match labels/<name>.rallies.csv <-> videos/<name>.<ext> (__main__.py:271-280).",
380         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:31,49",
381         "verdict": "correct",
382         "evidence": "backend/worker/__main__.py:332 def cmd_train; :359 lists
{corpus}/labels/; :383-384 fname.replace('.rallies.csv','') stem join. Logic present and
correct; cmd_train line drifted from :231 to :332.",
383         "suggestedFix": "Update cmd_train citation to __main__.py:332.",
384         "severity": "low"
385     },
386     {
387         "claim": "Reuse map: add a /own-model alias in site/server.js:49-51; Worker
env.ASSETS.fetch needs no change (site/src/index.js:84).",
388         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:52,73",
389         "verdict": "correct",
390         "evidence": "site/server.js:99 'pathname === \"/own-model\" ? \"/own-
model.html\"' and :41 LOCALIZED_PAGES includes own-model.html (alias shifted off the cited
:49-51). site/src/index.js:84 'return env.ASSETS.fetch(request)' with no /own-model branch ?
confirms the Worker needs no change.",
391         "suggestedFix": "Update the server.js alias citation (now ~:99, not
:49-51).",
392         "severity": "low"
393     },
394     {
395         "claim": "Reuse precedent: screenshots under site/public/img/ (precedent
site/og.png + make_og.py).",
396         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:52",
397         "verdict": "correct",
398         "evidence": "git ls-files: site/make_og.py and site/public/og.png both
exist; site/public/img/ holds annotate-enable.png, annotate-tool.png, studio-*.png. The og asset
path is site/public/og.png (the doc wrote 'site/og.png' loosely), make_og.py is at
site/make_og.py.",
399         "suggestedFix": "Minor: the OG image is at site/public/og.png (not
site/og.png).",
400         "severity": "low"
401     },
402     {
403         "claim": "SPA today: web/src/App.tsx is one static screen, no router
(web/package.json); a correction screen needs react-router-dom.",
404         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:53,64",
405         "verdict": "stale",
406         "evidence": "grep on web/package.json: no react-router dep (claim 'no
router' still TRUE). But App.tsx is no longer 'one static screen' ? it now imports and
conditionally renders SetupWizard (web/src/App.tsx:11,65,242, P2c first-run wizard from PR #88).
The 'one static screen' framing predates the wizard.",
407         "suggestedFix": "Reword to 'no router; App.tsx renders a first-run
SetupWizard + the ingest panel' so the SPA surface description matches the post-#88 code.",
408         "severity": "low"
409     },
410     {
411         "claim": "SPA correction route depends on a new, unbuilt engine endpoint
PATCH /api/segments/{id} (dead columns
candidate_segments.human_label/notes/confirmed_segment_id); P8 ? Gated.",
412         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:53,79",
413         "verdict": "correct",
414         "evidence": "grep across backend/ for 'api/segments'/'PATCH' on
candidate_segments finds only the candidate_segments table + query_candidate_segments readers
(database.py:217, eval/*.py); no PATCH /api/segments route exists. P8 is still genuinely unbuilt
as stated.",
415         "suggestedFix": "",
416         "severity": "low"
417     },
418     {
419         "claim": "Relation note: peer of LOCAL_APPLIANCE_ARCHITECTURE.md (the latter
lands with PR #13).",

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420         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:6,84",
421         "verdict": "stale",
422         "evidence": "git ls-files confirms docs/LOCAL_APPLIANCE_ARCHITECTURE.md
already exists in the repo; it has long since landed, so the forward-looking 'lands with PR #13'
/ 'cross-link once PR #13 merges' phrasing is outdated.",
423         "suggestedFix": "Drop the 'lands with PR #13' future-tense framing ? the doc
exists now; add the cross-link.",
424         "severity": "low"
425     },
426     {
427         "claim": "rally-annotator is a single-file VLC Lua extension, VERSION 1.6.4,
MIT-licensed, public at github.com/Khelsutra/rally-annotator.",
428         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:29",
429         "verdict": "correct",
430         "evidence": "rally-annotator/vlc/rally_annotator.lua:1 '(v1.6.4)'; LICENSE
first line 'MIT License'; own-model.html:109/191 links to https://github.com/Khelsutra/rally-
annotator (org-link migration was PR #23, captured in the doc's own history).",
431         "suggestedFix": "",
432         "severity": "low"
433     },
434     {
435         "claim": "$7 tracker: P1/P2/P3/P4 all ? via '(this PR)'.",
436         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:72-75",
437         "verdict": "correct",
438         "evidence": "own-model.html exists (P1); /own-model alias in server.js:99
(P2); nav cross-link index.html:255 + footer :620 (P3); verbatim annotator content + GitHub link
in own-model.html:108-191 (P4). All shipped ? git history shows these in PR #22 (b1110e1,
'annotate-train P1-P4'), so '(this PR)' should now read '#22'.",
439         "suggestedFix": "Replace the '(this PR)' placeholders in the PR column with
#22 (and #23 for the org-link follow-up) so the tracker cites real merged PRs.",
440         "severity": "medium"
441     }
442 ]
443 },
444 {
445     "doc": "docs/DRY_RUN_CUJ.md",
446     "overallStatus": "minor-staleness",
447     "summary": "The doc's substantive engineering claims are accurate and well-
grounded: the SPA back-half + B-P2 front-half ingest screen exist
(web/src/components/IngestPanel.tsx, web/src/hooks/useIngestJob.ts, #56), the engine bucket-
ingest routing is verified in code, and the three cited engine PRs (#280/#281/#289) plus all
four cited engine tests exist. The staleness is in metadata/framing: the header still says
Status: DRAFT though docs/README.md correctly calls it DELIVERED with DoD $6 boxes 1-3 done, and
the doc repeatedly describes the multilingual workflow as \"en/kn/hi\" (3 langs) when the SPA
now supports 6 (es/da/id added in #87). It is a completed tracker and an archive candidate once
the parked droplet box is closed.",
448     "findings": [
449         {
450             "claim": "Status: DRAFT ? owner-gated . ... Goal for this session.",
451             "docLocation": "docs/DRY_RUN_CUJ.md:3",
452             "verdict": "stale",
453             "evidence": "docs/README.md:4 marks this doc '? DELIVERED (2026-06-21)'; DoD
$6 boxes 1-3 are all checked [x] (DRY_RUN_CUJ.md:47-49) and the work shipped (#53/#55/#56/#57).
A DRAFT/'goal for this session' header no longer matches the completed, delivered state.",
454             "suggestedFix": "Flip the header to Status: DONE / DELIVERED (2026-06-21)
and drop 'Goal for this session'; note the only open box (droplet, $6 box 4) is owner-parked.
Consider archiving once that box closes.",
455             "severity": "medium"
456         },
457         {
458             "claim": "The SPA is 'Fully en/kn/hi (the switcher / ?lang=)' and the
multilingual workflow is en/kn/hi.",
459             "docLocation": "docs/DRY_RUN_CUJ.md:11",
460             "verdict": "stale",
461             "evidence": "web/src/i18n/index.ts:25 ? `export const SUPPORTED_LOCALES =

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[\"en\\\", \"kn\\\", \"hi\\\", \"es\\\", \"da\\\", \"id\\\"]`; es/da/id were added to the SPA in PR #87
(cf59fbf 'add es/da/id to the marketing site (parity with the SPA's 6 languages)'). The SPA now
supports 6 languages, not 3.",
462     "suggestedFix": "Update 'en/kn/hi' to 'en/kn/hi/es/da/id (6 languages)'
where it describes the current switcher; the Hindi/Kannada run results in §7 stay accurate as
historical record.",
463     "severity": "low"
464 },
465 {
466     "claim": "This SPA front-half (ingest/process UI) is the *only* net-new work
for Path B (= B-P2); the SPA has no ingest/process UI today.",
467     "docLocation": "docs/DRY_RUN_CUJ.md:14",
468     "verdict": "correct",
469     "evidence": "B-P2 was net-new work at draft time and has since shipped:
web/src/components/IngestPanel.tsx + web/src/hooks/useIngestJob.ts (first landed in PR #56
'feat(web): B-P2 ? in-UI bucket ingest', 4525c5d, 2026-06-21). useIngestJob.ts drives POST
/api/process (processVideo) ? poll GET /api/jobs/{id} (getJob) ? hand video_id to App, exactly
as scoped.",
470     "suggestedFix": "",
471     "severity": "low"
472 },
473 {
474     "claim": "POST /api/process accepts gs://gcs://s3://local:// URIs and skips
the local-existence check for non-local backends; resolve via storage.resolve_input_ref;
gs://?gcs normalization.",
475     "docLocation": "docs/DRY_RUN_CUJ.md:15,68-72",
476     "verdict": "correct",
477     "evidence": "Engine backend/main.py:581 `input_ref =
storage.resolve_input_ref(req.video_path, ...)` and main.py:590 `if input_ref.backend ==
'local' and not os.path.exists(input_ref.key):` (the non-local skip), with the fetch at
main.py:458 `storage.acquire_local_input(...)`; backend/storage/ref.py:60 `if scheme == 'gs':
scheme = 'gcs'`. Substance matches; the exact line numbers 159/274/283 cited in the doc have
drifted as the engine shipped further PRs.",
478     "suggestedFix": "Optionally refresh the cited line numbers (now
~main.py:581/590/458, ref.py:60) since the engine surface moved; the routing logic itself is
intact.",
479     "severity": "low"
480 },
481 {
482     "claim": "Tested by
tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches,
::test_process_hosted_mode_rejects_nonlocal_uri, ::test_process_unknown_scheme_returns_400,
::test_cli_nonlocal_input_fetches_and_keeps_uri.",
483     "docLocation": "docs/DRY_RUN_CUJ.md:15,76",
484     "verdict": "correct",
485     "evidence": "Engine repo grep: tests/test_api_endpoints.py:508
test_process_nonlocal_uri_skips_exists_and_fetches; :577
test_process_unknown_scheme_returns_400; :608 test_process_hosted_mode_rejects_nonlocal_uri;
:627 test_cli_nonlocal_input_fetches_and_keeps_uri. All four exist with the asserted names.",
486     "suggestedFix": "",
487     "severity": "low"
488 },
489 {
490     "claim": "B-P1 (gs://?/api/process?API-DB video_id) already shipped in
engine M2 / #280; the job-drain hang fixed in #281; the compile-from-bucket 500 fixed in #289.",
491     "docLocation": "docs/DRY_RUN_CUJ.md:15,63,68,88",
492     "verdict": "correct",
493     "evidence": "Engine git log: e111add 'feat(serving): M2 ? URI-aware input
... (#280)'; 41d45fd 'fix(server): drain async jobs under `python -m backend.main` (double-
module hang) (#281)'; e68b583 'fix(api): resolve a bucket-ingested video's source to a local
file before stitching (#289)'. All three PRs exist with descriptions matching the doc.",
494     "suggestedFix": "",
495     "severity": "low"
496 },
497 {

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498         "claim": "DoD §6 box 3: in-UI Path-B bucket?reel CUJ DONE (live e2e
2026-06-21) ? 12 rallies loaded ? reel downloaded; box 4 (CPU droplet) still open/un-
provisioned.",
499         "docLocation": "docs/DRY_RUN_CUJ.md:49-50",
500         "verdict": "correct",
501         "evidence": "Box 3 is corroborated by PR #57 'docs(dry-run): Path-B live e2e
DONE ? in-UI bucket?reel CUJ + screencast (DoD §6 box 3 ?)' and the merged B-P2 SPA (#56).
docs/README.md:6-7 confirms box 4 (droplet, option 3) remains 'parked by the owner (GPU
subscription)'. The run-result specifics (12 rallies, reel sizes, GCS screencast paths) are
runtime artifacts not locally reproducible.",
502         "suggestedFix": "",
503         "severity": "low"
504     },
505     {
506         "claim": "Run-result specifics: 12 rallies / video_id 05e0e577 /
50.9MB+76.5s reels / screencasts at gs://khelsutra-rally-corpus/promo/*.mp4 and the staged
bucket video gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4.",
507         "docLocation": "docs/DRY_RUN_CUJ.md:54-60,82-85",
508         "verdict": "unverifiable",
509         "evidence": "These are GCS bucket objects and a one-off local run's outputs;
they cannot be checked from the repo. No counter-evidence found ? the surrounding code/PR claims
that frame them are all verified.",
510         "suggestedFix": "",
511         "severity": "low"
512     }
513 ]
514 },
515 {
516     "doc": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md",
517     "overallStatus": "significant-staleness",
518     "summary": "A still-active DRAFT tracker (last updated 2026-06-21) whose §9
progress and several \"Absent (net-new)\" engine-fact anchors have drifted: the engine has since
shipped /api/capabilities, exposure-safety boot posture + an edge-auth layer, and merged
native_wasb_runner to master, while the alpha deploy went LIVE on 2026-06-22
(app.khelsutra.guru). The khelsutra-side claims (PR refs #11/#56/#64/#66, IngestPanel,
VITE_API_BASE) check out; the staleness is concentrated in engine status and the deploy/DoD
progress. Not yet an archive candidate (open owner gates remain) but needs a refresh pass.",
519     "findings": [
520     {
521         "claim": "§3 lists `GET /api/capabilities` under 'Absent (net-new): ...
`[verified ? grep returns none]`; §4.6 Reuse map lists it under 'Net-new'; §9 T7 (capabilities
+ health + banner) is ? Todo.",
522         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:49,113,164",
523         "verdict": "stale",
524         "evidence": "engine origin/master backend/main.py:340
`@app.get(\"/api/capabilities\")` (and main.py:307 `/api/health`, main.py:334 `/api/version`)
all exist on master. git grep '/api/capabilities' origin/master returns a hit.",
525         "suggestedFix": "Move `/api/capabilities` out of 'Absent (net-new)' and out
of §4.6 'Net-new'; mark the engine half of T7 (capabilities+health) as shipped (?/?), leaving
only the SPA banner pending.",
526         "severity": "high"
527     },
528     {
529         "claim": "TL;DR + Risk table: 'the engine has zero app-layer auth and CORS
`*` `[verified backend/main.py:42]` ? structural safety only because the engine binds
127.0.0.1.",
530         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:14,30,103,128",
531         "verdict": "stale",
532         "evidence": "engine origin/master backend/main.py:82 uses
`allow_origins=cors_origins_from_env()` (default ['*'] but env-lockable) with
`allow_credentials=False`; main.py:109 adds TrustedHostMiddleware (DNS-rebinding/Host
allowlist); backend/api/auth.py + config/startup.py enforce_startup_checks + M9 `Cf-Access-
Authenticated-User-Email` `owner_id` now exist on master. CORS is no longer a hardcoded bare
`*`, and an app-layer edge-auth integration now exists.",
533         "suggestedFix": "Reword to: CORS default is `*` but now env-configurable

```

```

(allow_credentials off), and the engine now ships a Host-header guard + an edge-auth integration
(Cf-Access trust) + exposure-safety boot posture ? so 'zero app-layer auth' is no longer
literally true. Re-anchor the line (main.py:42 no longer points at CORS).",
534         "severity": "high"
535     },
536     {
537         "claim": "$3 / $4.3: 'Native WASB is in-flight on branch `feat/native-wasb-
runner`: `native_wasb_runner.py` exists `[verified]`'.",
538         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:45,96",
539         "verdict": "stale",
540         "evidence": "engine origin/master now contains
backend/pipeline/detectors/native_wasb_runner.py and tests/test_native_wasb_runner.py (git ls-
tree origin/master). The runner has merged to master; the `feat/native-wasb-runner` branch still
exists but the file is no longer branch-only. (trajectory_hybrid._build_native_runner still
raises NotImplementedError at trajectory_hybrid.py:61 ? that sub-claim is still correct.)",
541         "suggestedFix": "Change 'in-flight on branch feat/native-wasb-runner' to
'native_wasb_runner.py is on master; trajectory_hybrid still refuses, fusion builds it; owner
byte-parity gate still pending.'",
542         "severity": "medium"
543     },
544     {
545         "claim": "$9 alpha-deploy block: 'the actual stand-up (tunnel + Pages + CF
Access + key rotation) is the **owner** step'; $11 DoD box 1 (CUJ-1 e2e on droplet behind gate)
and box 2 (raw :8000 refused, tested) unchecked.",
546         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:175,182,183",
547         "verdict": "stale",
548         "evidence": "deploy/droplet/README.md:100-106 '? Live-verified on the
droplet (2026-06-22)': engine booted on 168.144.126.176; suite 1313 passed; boot-posture fail-
fast, Host-guard 400?200, path-jail, T4 upload?jail, and edge-auth 401-gate all exercised live;
full DNS?Cloudflare?Access?tunnel?engine chain confirmed; app.khelsutra.guru up. Only Gemini-key
rotation + a browser OTP login remain.",
549         "suggestedFix": "Update $9 to note the alpha stand-up is LIVE (2026-06-22,
app.khelsutra.guru, posture test exercised) with only key rotation + browser-OTP login
outstanding; tick/qualify the corresponding DoD boxes.",
550         "severity": "high"
551     },
552     {
553         "claim": "Status header: 'Status: DRAFT ... Last updated: 2026-06-21' with
$9 the source of truth.",
554         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:3",
555         "verdict": "stale",
556         "evidence": "khelsutra git log shows merges through #94 (master HEAD
50b2bd7) and the alpha went live 2026-06-22, both after the doc's 2026-06-21 'last updated'.
Multiple $9 rows and engine facts have moved since (T7 engine half, alpha deploy, native_wasb on
master).",
557         "suggestedFix": "Bump 'Last updated' and consider promoting DRAFT?IN
PROGRESS; refresh $9 rows to current reality.",
558         "severity": "medium"
559     },
560     {
561         "claim": "D4 'Embed RallyPicker ... PR [#11]' and $9 T4 '[#66]', T12 'SPA
#56', alpha-deploy '[#64]' / 'VITE_API_BASE'.",
562         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:33,161,169,175",
563         "verdict": "correct",
564         "evidence": "khelsutra git log: cb74ce5 '...RallyPicker (#11)'; ca619fb 'T4
? in-UI file upload (#66)'; 4525c5d 'B-P2 ? in-UI bucket ingest (#56)'; 8bbc557 'VITE_API_BASE
(#64)'. web/src/components/IngestPanel.tsx handles both chunked upload (T4) and a gs:// URI
(T12); web/src/api/client.ts:25 reads VITE_API_BASE.",
565         "suggestedFix": "",
566         "severity": "low"
567     },
568     {
569         "claim": "T12 (bucket-URI ingest) marked ? : 'engine accepts URIs (M2/#280) +
SPA ingest screen shipped (B-P2, #56); in-UI gs:///reel CUJ proven live e2e (DRY_RUN_CUJ.md
$9)'.",

```

```

570         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:169",
571         "verdict": "correct",
572         "evidence": "docs/DRY_RUN_CUJ.md:80-92 §9 records the live in-UI gs://?reel
CUJ (12 rallies ? reel downloaded) and the compile-from-bucket 500 fix (engine #289);
IngestPanel.tsx accepts a gs:// URI. khelsutra #56 (B-P2) is merged.",
573         "suggestedFix": "",
574         "severity": "low"
575     },
576     {
577         "claim": "T11 retention marked ?: sweep tool shipped (engine #299), auto-
scheduling tracked (engine #301).",
578         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:168,130",
579         "verdict": "unverifiable",
580         "evidence": "These reference the engine repo's GitHub PR #299 / issue #301
(Khelsutra/badminton-highlight-indexer). Not falsifiable from the local khelsutra checkout; the
local engine checkout is on a feature branch and I did not confirm the specific sweep tool by PR
number. The khelsutra-side row metadata is internally consistent (#69 flipped T11 to ? per git
log bea8a2a).",
581         "suggestedFix": "",
582         "severity": "low"
583     },
584     {
585         "claim": "§3 Droplet reality: co-hosts the marketing site at
`/srv/khelsutra-site`, node `:8787`, engine at `~/rally`; MinIO 9000/9001 + GCS emulator 9023.",
586         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:48",
587         "verdict": "unverifiable",
588         "evidence": "These are live-droplet filesystem/port facts (168.144.126.176)
not checkable from the local repo. deploy/DEPLOY_ALPHA.md:31 and deploy/droplet/README.md:4
corroborate the IP/CPU-no-GPU/blr1 framing, but the exact paths/ports (/srv/khelsutra-site,
:8787, ~/rally) are not in-repo verifiable.",
589         "suggestedFix": "",
590         "severity": "low"
591     },

```

4. user (2026-06-22T09:46:29.465Z)

```

1      # Annotate your footage ? train your own model (KhelSutra Studio USP pages)
2
3      > **Status:** `DRAFT` ? owner-gated on scope . **Owner:** Avi . **Created:** 2026-06-20
. **Last updated:** 2026-06-20
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5      > **Tracking anchors:** §7 progress tracker is the source of truth; index in
`docs/README.md`; pointer in `NEXT_STEPS.md` + memory `current-state` once work starts.
6      > **Relation:** peer of `khelsutra/docs/PER_RALLY_SELECTION.md` and
`khelsutra/docs/LOCAL_APPLIANCE_ARCHITECTURE.md` (the latter lands with PR #13); mirrors the
engine repo's `PROJECT_DOC_TEMPLATE` discipline. Documents an EXTERNAL public tool (`rally-
annotator`, MIT) and the engine's `backend/worker train` CLI; it does not own those repos.
7      > **Honesty note:** every engine-fact claim is tagged `[verified]` (read against code
this session), `[design]` (proposed here), or `[researched]` (needs sign-off). Not legal advice.
8
9      ---
10
11     ## 0. TL;DR
12     Ship ONE new static **marketing page** (`site/public/own-model.html`, slug `~/own-model`)
with three anchored sections ? **Annotate your footage . Train your own model . Setup & use** ?
telling the bring-your-own-model story and linking the real, public, MIT **rally-annotator** VLC
plugin `[verified]`. The single most important caveat: only the **ANNOTATE half is real today**
? the rally-annotator ships (v1.6.4, MIT) and its CSV is directly ingestible by the engine with
**no transform** `[verified]` ? while the **TRAIN half is an open, in-build CLI** whose GPU-free
path trains a **pipeline-validation mock only**, and whose **real model needs a GPU + an in-
flight native WASB runner** behind an **owner-gated ship decision** `[verified gate_verdict
harness.py:98]`. The Studio SPA gets a **design-tracked, owner-gated** correction/contribute
screen, **never a "Train now" button**. Defensible claim = **ownership / licensing / privacy /
consent / minors-excluded**, NOT accuracy-superiority (there is no benchmark vs a paid LLM).

```

```

13
14     ## 1. Problem & goal
15     KhelSutra's differentiator vs paid LLMs is that **you can own a model trained on YOUR
footage1 ? your courts, your camera angles, your busy multi-court halls ? on a commercially-
clean, inspectable stack. The concrete, usable entry point already exists: **rally-annotator2,
a free MIT VLC Lua extension that turns "watch a match" into "produce golden labels" 3[verified
README.md:22-24]4. What's missing is the **education/positioning surface5 that tells this story
and the **honest documentation6 of the end-to-end loop. "Good" = a viewer understands the moat,
can install the annotator and produce labels today, and is **never misled7 into thinking one-
click training exists. Read first: `rally-annotator/README.md`, `rally-
annotator/docs/CSV_FORMAT.md`, `badminton-highlight-indexer/backend/worker/__main__.py`,
`khelsutra/docs/PER_RALLY_SELECTION.md`, `khelsutra/site/public/index.html` (the honesty voice).
16
17     ## 2. Decisions locked
18     | # | Decision | Source / date | Implication |
19     |---|-----|-----|-----|
20     | D1 | Marketing page ships NOW; SPA screen is design-tracked + owner-gated | This
synthesis 2026-06-20 | Convert surface first; operate surface deferred |
21     | D2 | ONE marketing page, three anchored sections (Annotate/Train/Setup) | Avoid thin
pages; annotator is external | Clone `capture.html` skeleton; no build step |
22     | D3 | Annotator docs live on marketing; SPA only links out | `[verified]` annotator is
a standalone VLC tool | Do not re-author VLC instructions in the SPA |
23     | D4 | NO "Train now" button/endpoint in the SPA | `[verified]` training is offline
`backend/worker train` | SPA's honest role = correction loop ? grow corpus |
24     | D5 | Frame USP as ownership/licensing/privacy/consent, NOT accuracy | `[verified]` no
user-model-vs-LLM benchmark exists | No accuracy-superiority copy anywhere |
25     | D6 | Paid LLMs never a training source; MIT/Apache/BSD only; minors excluded; opt-in |
`[verified CLAUDE.md]` + `harness.py:178,189-192` attestation | Reuse the site's existing minors
+ opt-in wording (`index.html:342-349`) |
26     | D7 | Annotator CSV needs NO transform for the engine label path | `[verified]` $4.2
schema bridge | Document the MD5+stem join as the only real prereq |
27
28     ## 3. Foundation ? research / prior art `[verified this session]`
29     **The annotator (real, shipping).8 **rally-annotator9 is a single-file VLC **Lua
extension10 (`vlc/rally_annotator.lua`, VERSION 1.6.4), MIT-licensed, public at
https://github.com/Khelsutra/rally-annotator. It is button-driven (no listener capabilities;
deliberate ? VLC pause-listener is flaky on macOS / broken in 4.0), reads playback time via
`vlc.var.get(input, 'time')` (micros/1e6?s, lines 201-207), and appends one row per rally to
`<video-stem>.rallies.csv` next to the video with atomic rewrite (save_all 310-335). Targets 5
net-separated racquet sports sharing one error taxonomy. CI runs a Lua 5.1 dialog suite +
layout-snapshot golden diff. VLC **3.0.x only11 (4.0 changed the Lua API).
30
31     **The train loop (partly real).12 The engine's `backend/worker train` (cmd_train,
`__main__.py:231`) pulls `labels/` (+ `videos/` for the detector source), builds per-video
trajectory CSVs, assembles a manifest, **shells out13 to `python -m training.gen0.harness` (the
one-way train/serve wall, `training/README.md:9-11`), and pushes a versioned bundle to
`weights/<version>/`. Two trajectory sources, identical downstream: **synth14 (DEFAULT, GPU-
free, a CPU mock that ENCODES the labels ? plumbing-meaningful only,
`stub_runner.py:124-149,131-133`) and **detector15 (real WASB, M3.5, **GPU-gated, IN-FLIGHT16 ?
the Linux-native runner does not exist yet, `02:15`, `04:98-113`). The harness fits a ~1M-param
LogisticRegression on CPU and emits the artifact bundle (`weights.json` + reviewable
`manifest.json`); **serving consumes only the bundle, never a code import17 (ADR-008). The
**ship-gate18 (`gate_verdict`, `harness.py:98`) currently refuses a commercial ship19 regardless
of F1: WASB checkpoint provenance C3 unresolved (commercial ship refused) + F1@tIoU not yet
calibrated (#172).
32
33     **Guardrails (verbatim, non-negotiable).20 Paid LLMs (Gemini/OpenAI/Anthropic) =
inference/serving/fallback ONLY, **never a training data source21 (bundle attests
`no_paid_llm_training_labels:True`, `harness.py:192`); label source must be human (annotator
golden CSVs, `harness.py:178`). Commercial-clean deps only ? **MIT/Apache-2.0/BSD22. Owned base
= Qwen3-VL (Apache) primary; Gemma-4 AMBER/bench-only; Gemma 3 rejected. **Minors excluded23
from the training pool; per-user training data needs **explicit opt-in24 `[verified CLAUDE.md]`.
34
35     ## 4. Design / architecture
36

```

```

37     ### 4.1 The annotate ? CSV ? train ? model flow `[verified except where noted]`
38     1. **ANNOTATE** (rally-annotator, VLC): per rally, Mark START (serve-contact) ? Mark END
(shuttle-dead) ? Ending reason ? optional shots ? Save Rally ? appends a row to
`<video>.rallies.csv`. No compute, no GPU, no LLM.
39     2. **STAGE**: `python -m src.cli.curate import-local --sport badminton --video-path
match.mp4 --annotations-path match.rallies.csv`, and/or populate a corpus prefix `videos/
labels/ manifest.json weights/ outputs/<video_id>/`.
40     3. **TRAIN**: default `python -m backend.worker train --corpus-uri ? --out-uri ?
--version 0.1.0` (synth, GPU-free CPU mock, ?2 labeled videos); real = add `--trajectory-source
detector --segmenter wasb_hybrid` with `indexing.detector_impl=native` (GPU + native WASB +
`wasb_badminton_best.pth.tar`, IN-FLIGHT).
41     4. **HARNESS ? BUNDLE**: parse trajectory + load GT ? frame features ? honest LOVO eval
? fit final LR + tune threshold ? ship-gate ? emit `weights.json` + `manifest.json`.
42     5. **USE** `[researched/design, owner-gated]`: if Gen-0 clears the ship-gate A/B, the
bundle drives the existing record?upload?reel loop. No per-user serving UI; no Studio Train
button.
43
44     ### 4.2 The schema bridge ? annotator CSV ? engine label CSV (MATCHES, no transform)
45     Annotator writes header
`rally_number,start_time,end_time,ending_reason,sport,shots_count` (`rally_annotator.lua:128`;
`CSV_FORMAT.md:6`), `%3f` decimal seconds, end>start. The engine reads **by column name** via
two readers, both satisfied:
46     - **Default synth path** ? `synth_trajectory_from_labels` uses `csv.DictReader` reading
literally `row["start_time"]`/`row["end_time"]` (`stub_runner.py:139`) ? **MATCHES exactly**;
extra columns ignored. (This path needs the `*_time` names specifically ? the short `start/end`
alias is NOT accepted here.)
47     - **Canonical GT reader** ? `load_gt_intervals` accepts `start`/`end` OR
`start_time`/`end_time`, ignores other columns, also accepts headerless positional col0,col1,
tolerates decimal-seconds or MM:SS/HH:MM:SS, BOM, blank/`#` lines (`gt_loader.py:14-90`) ?
**MATCHES** the `*_time` branch.
48
49     The worker's own inference CSV uses `start,end,shot_count,ending_reason`
(`__main__.py:62-74`) ? order/short-name differ, but order never matters and `gt_loader` accepts
both aliases; the annotator's `*_time` form is the **safer** of the two. **The ONE real
prerequisite is an IDENTITY/JOIN constraint, not a schema gap:** training keys video?label by
**file MD5** and stem-matches `labels/<name>.rallies.csv` ? `videos/<name>.<ext>`
(`__main__.py:271-280`; `02:48-60`). Stems must agree and the staged video bytes must be **byte-
identical** to what was annotated ? any re-encode changes the MD5 and breaks the join (collector
manifest currently omits `md5`, M2 prereq, `02:62-68`).
50
51     ### 4.3 Reuse map
52     - **Marketing** `[verified]`: clone `site/public/capture.html` skeleton + brand tokens
(`index.html:25-32`); reuse `.honest`/`.notes` components (`:152-158`, used `:351-354`,
`:400-415`); add a `own-model` alias in `site/server.js:49-51` (Worker `env.ASSETS.fetch` needs
no change, `site/src/index.js:84`); cross-link nav `:218-223`, the "gets sharper" section
`:327-356`, footer `:528`; screenshots under `site/public/img/` (precedent `site/og.png` +
`make_og.py`).
53     - **SPA** `[design, gated]`: today `web/src/App.tsx` is one static screen, **no router**
(`web/package.json`). A correction screen needs react-router-dom (or a hash switch) + the typed
client `web/src/api/client.ts`/`types.ts` + `web/src/lib/log.ts`; it realizes P10 in
`PER_RALLY_SELECTION.md:122` and depends on a **new** engine endpoint `PATCH /api/segments/{id}`
(dead columns `candidate_segments.human_label/notes/confirmed_segment_id`).
54     - **Annotator content** `[verified]`: embed CSV example + install commands + ending-
reason table **verbatim** from the repo; link, don't re-author.
55
56     ## 5. Threat model / privacy
57     - **Footage stays the user's**: the annotator runs locally in the user's own VLC and
writes a CSV next to the video ? labeling happens with **footage never leaving the machine**
`[verified]`.
58     - **Training use = explicit opt-in, never automatic** `[verified CLAUDE.md]`; reuse the
site's existing opt-in wording.
59     - **Minors excluded from any training pool** `[verified CLAUDE.md + live site "Minors
are protected"]` ? keep prominent (audience = academies with children).
60     - **No paid-LLM laundering**: the model never trains on a paid LLM's output
(terms/copyright), attested in the bundle (`harness.py:192`).

```

```

61      - **Residual**: base-model (Qwen3-VL) pretraining provenance is unauditale
`[researched]` ? owner risk-acceptance required before VLM "train your own model" copy goes
beyond roadmap.
62
63      ## 6. Honest limits ? what this does NOT do
64      A shipped marketing page does NOT mean training is one-click (it is an offline CLI),
does NOT mean a real model exists (synth = mock; real = GPU + in-flight native WASB + owner-
gated ship-gate), and does NOT imply any accuracy claim vs a paid LLM (no benchmark exists). The
annotator is a **separate VLC 3.0.x desktop plugin**, not in-browser; it is **click-driven** (no
marking hotkeys). The SPA correction screen is **design-tracked + owner-gated** and depends on
an unbuilt `PATCH /api/segments` endpoint. Two CSV schemas exist (annotator lighter vs Roora
vendor) ? do not conflate.
65
66      ## 7. Deliverables & progress tracker ? source of truth
67      Legend: ? Todo · ? In progress · ? Done · ? Gated. One small PR per row, branched from
`origin/master`, no stacking.
68
69      | ID | Deliverable | Depends on | Gated? | Status | PR |
70      |----|-----|-----|-----|-----|----|
71      | P0 | This tracked doc in `khelsutra/docs/`, indexed in `docs/README.md` | ? | No | ? |
[#15](https://github.com/avidullu/khelsutra/pull/15) |
72      | P1 | `site/public/own-model.html` (clone capture.html; 3 sections; honesty fences;
placeholder shots) | P0 | No | ? | (this PR) |
73      | P2 | `own-model` alias in `site/server.js` (Worker needs no change) | P1 | No | ? |
(this PR) |
74      | P3 | Cross-links: nav + footer (the "gets sharper" section link is a nice-to-have,
deferred) | P1 | No | ? | (this PR) |
75      | P4 | Embed verbatim repo text (CSV/install/ending-reasons) + GitHub link | P1 | No | ?
| (this PR) |
76      | P5 | Capture + commit real screenshots (consented, no minors, faces redacted); replace
placeholders | P1 | ? Owner (footage) | ? | ? |
77      | P6 | Owner: slug + 1-vs-3-page + train-positioning honesty | ? | ? Owner | ? | ? |
78      | P7 | [design] SPA correction/contribute route (RallyReview + link-out) | P8 | ? Owner
| ? | ? |
79      | P8 | [design] Engine `PATCH /api/segments/{id}` (M4 corpus extension) | ? | ? Owner
(engine) | ? | ? |
80      | P9 | [researched] Serve a shipped Gen-0 bundle into record?reel | native WASB + ship-
gate | ? Owner+blocked | ? | ? |
81
82      ## 8. Open questions ? owner / external
83      **Blocking gates:** (1) Greenlight how prominently to position "train your own model"
(CLI + roadmap). (2) Approve the SPA correction screen + the new `PATCH /api/segments/{id}`
endpoint. (3) Owner risk-acceptance on Qwen3-VL provenance before VLM copy. (4) Provide
consented no-minors footage for the 5 screenshots.
84      **Nice-to-knows:** slug (`own-model` vs `byo-model` vs `annotate`); one page vs
three; react-router-dom vs hash switch; cross-link `docs/LOCAL_APPLIANCE_ARCHITECTURE.md` once
PR #13 merges.
85
86      ## 9. Definition of done
87      - [ ] `own-model.html` live with Annotate/Train/Setup sections, honesty fences, verbatim
repo text, GitHub link, real (non-fabricated) screenshots.
88      - [ ] `own-model` alias works on both the Node host and the Cloudflare Worker; cross-
linked from nav/section/footer.
89      - [ ] Every claim tagged `[verified]`/`[design]`/`[researched]`; no accuracy-superiority
claim; no "Train now" button; VLC-3.0.x-only + GPU/native-WASB-gated + owner-gated stated
plainly.
90      - [ ] Schema-bridge + MD5/stem-join prerequisite documented; the two CSV schemas kept
distinct.
91      - [ ] $7 tracker reflects real shipped state; doc indexed in `docs/README.md`.
92
93      ## 10. References
94      **Annotator `[verified]`: `rally-annotator/README.md`, `vlc/rally_annotator.lua`
(HEADER :128; now_seconds :201-207; marks :608-628; save_all :310-335; resolve_out_path
:253-267), `docs/CSV_FORMAT.md`, `docs/ENDING_REASONS.md`, `LICENSE` (MIT),
`.github/workflows/ci.yml`.

```

```

95     **Engine `[verified]`:** `backend/worker/__main__.py` (cmd_train :231;
_write_intervals_csv :62-74; join :271-280; harness shell-out :321-328),
`backend/pipeline/detectors/stub_runner.py` (synth :124-149, :131-133, :139),
`training/gen0/harness.py` (gate_verdict :98; label_source :178; attestation :189-192),
`backend/eval/gt_loader.py` (:14-90), `docs/SETUP_NEW_MACHINE.md`,
`docs/DECOUPLED_COMPUTE/02-COMPUTE-MAP-AND-CONTRACT.md`, `docs/DECOUPLED_COMPUTE/04-BUILD-AND-
TEST-PLAN.md`, `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`, `CLAUDE.md`.
96     **KhelSutra `[verified]`:** `site/public/index.html`, `site/public/capture.html`,
`site/server.js`, `site/src/index.js`, `web/src/App.tsx`, `web/package.json`, `docs/README.md`,
`docs/PER_RALLY_SELECTION.md`.
97
98     ### Changelog
99     - 2026-06-20 ? Initial DRAFT consolidating annotator + train-loop + UI-surface research
into one tracked doc (multi-agent design pass; schema-bridge + no-images + train-guardrails re-
verified against source before writing).
100

```

5. assistant (2026-06-22T09:46:40.074Z)

My worklist is the staleFindings for `docs/ANNOTATE\_AND\_TRAIN.md`. Let me extract those entries from the perDoc findings I've read. The findings with verdict "stale" for this doc are:

1. Status DRAFT / Last updated 2026-06-20 (line 3) ? medium
2. P5 Todo though annotator screenshots committed (line 76) ? high
3. Doc claimed indexed in docs/README.md but not (lines 5, 71, 91) ? high
4. SPA "one static screen" framing predates wizard (lines 53, 64) ? low
5. "lands with PR #13" future-tense (lines 6, 84) ? low
6. §7 "(this PR)" placeholders should cite #22/#23 (lines 72-75) ? medium

Plus low-severity line-citation drifts (correct facts, drifted citations):

- gate_verdict harness.py:98 ? :142 (lines 12, 31)
- no_paid_llm_training_labels :192?:236, label_source :178?:222 (lines 33, 60)
- cmd_train __main__.py:231 ? :332 (lines 31, 49)
- server.js alias :49-51 ? ~:99 (lines 52, 73)
- OG image site/og.png ? site/public/og.png (line 52)

Let me re-verify these against the live repos before editing. Let me start with the engine-side claims and the khelsutra-side claims in parallel.

6. user (2026-06-22T09:46:46.805Z)

```

=== gate_verdict ===
origin/master:training/gen0/harness.py:142:def gate_verdict(eval_res: dict, floor:
Optional[dict]) -> dict:
=== ship false / C3 / 172 ===
origin/master:training/gen0/harness.py:174:     reasons.append("ship-gate F1@tIoU target NOT YET
CALIBRATED ? provisional; see issue #172")
origin/master:training/gen0/harness.py:175:     reasons.append("WASB checkpoint provenance C3
UNRESOLVED ? commercial ship blocked regardless of F1")
origin/master:training/gen0/harness.py:177:     return {"ship": False, "floor_passed":
floor_passed, "sign_test": sign, "blockers": reasons}
=== no_paid_llm + label_source ===
origin/master:training/gen0/harness.py:222:         "label_source": "human (rally-annotator
golden CSVs)",
origin/master:training/gen0/harness.py:236:         "no_paid_llm_training_labels": True,
=== cmd_train + join ===
origin/master:backend/worker/__main__.py:332:def cmd_train(args: argparse.Namespace) -> int:
origin/master:backend/worker/__main__.py:383:         fname = storage.parse_uri(uri).name
# e.g. GX020094_proxy.rallies.csv
origin/master:backend/worker/__main__.py:384:         name = fname.replace(".rallies.csv",
""").replace(".csv", "")
=== synth ===
origin/master:backend/pipeline/detectors/stub_runner.py:124:def
synth_trajectory_from_labels(labels_csv: str, out_csv: str, *, fps: float = 30.0,
origin/master:backend/pipeline/detectors/stub_runner.py:137:         for row in

```

```
_csv.DictReader(f):
```

7. user (2026-06-22T09:46:50.059Z)

```
=== own-model alias in server.js ===
origin/master:site/server.js:41:const LOCALIZED_PAGES = new Set(["index.html", "capture.html",
"own-model.html"]);
origin/master:site/server.js:99:      : pathname === "/own-model" ? "own-model.html"
=== LOCALIZED_PAGES ===
origin/master:site/server.js:41:const LOCALIZED_PAGES = new Set(["index.html", "capture.html",
"own-model.html"]);
origin/master:site/server.js:109:      if (LOCALIZED_PAGES.has(rel)) {
=== og.png ===
site/make_og.py
site/public/og.png
=== annotate screenshots ===
site/public/img/annotate-enable.png
site/public/img/annotate-tool.png
=== docs/README index ===
=== LOCAL_APPLIANCE exists ===
docs/LOCAL_APPLIANCE_ARCHITECTURE.md
```

8. assistant (2026-06-22T09:46:53.658Z)

All findings confirmed. Let me verify the remaining details: the `\_\_main\_\_.py:271-280` join citation, stub_runner exact line for `row['start\_time']`, the App.tsx SetupWizard, and the PR history for #22/#23/#25 + the own-model.html screenshot references.

9. user (2026-06-22T09:46:56.421Z)

```
=== stub_runner start_time line ===
origin/master:backend/pipeline/detectors/stub_runner.py:139:
rallies.append((float(row["start_time"]), float(row["end_time"])))
=== harness shell-out ===
origin/master:backend/worker/__main__.py:72:def _write_intervals_csv(segments: List[dict], path:
str) -> None:
origin/master:backend/worker/__main__.py:240:      _write_intervals_csv(segments,
os.path.join(out_local, "intervals.csv"))
origin/master:backend/worker/__main__.py:332:def cmd_train(args: argparse.Namespace) -> int:
origin/master:backend/worker/__main__.py:334:      to training.gen0.harness (backend must NOT
import training) -> push the artifact bundle.
origin/master:backend/worker/__main__.py:424:      cmd = [sys.executable, "-m",
"training.gen0.harness",
=== gt_loader ===
origin/master:backend/eval/gt_loader.py:30:_START_NAMES = ("start", "start_time")
origin/master:backend/eval/gt_loader.py:37:      start_idx = next((i for i, c in enumerate(cells)
if c in _START_NAMES), None)
origin/master:backend/eval/gt_loader.py:44:def load_gt_intervals(csv_path: str, *, strict: bool
= True) -> List[Interval]:
```

10. user (2026-06-22T09:46:59.478Z)

```
=== App.tsx SetupWizard ===
origin/master:web/src/App.tsx:11:import { SetupWizard, isWizardDone } from
"./components/SetupWizard";
origin/master:web/src/App.tsx:242:      {!videoId && !wizardDone && <SetupWizard onClose={()
=> setWizardDone(true)} />}
=== PR history #22 #23 #25 #15 ===
17242a9 feat(site): localize own-model body content to kn/hi (#86)
26d636d feat(site): link the academy flyer from the top nav + own-model page (#44)
9f0942b feat(site): own-model ? add an obvious back-to-home link; backlog: sister-repo org move
(#26)
1906c82 feat(site): own-model ? gate the reel, show the real annotator screenshots + a "default
works" note (#25)
```

f09e348 feat(site): reuse the sample reel on the own-model page (click-to-play, privacy-consistent) (#24)
87520b8 chore(site): point rally-annotator links at the new Khelsutra org (#23)
b1110e1 feat(site): own-model page ? annotate your footage & train your own model (annotate-train P1-P4) (#22)
6b62ebf docs(annotate-train): plan rally-annotator integration + "train your own model" USP pages (#15)
=== own-model.html screenshots ===
origin/master:site/public/own-model.html:111: <figure><figcaption data-i18n="site.ownModel.annotate.cap1">Enable it once: VLC ? View ? Rally Annotator.</figcaption></figure>
origin/master:site/public/own-model.html:112: <figure><figcaption data-i18n="site.ownModel.annotate.cap2">Mark each rally's start/end + ending reason, then Save Rally.</figcaption></figure>

11. assistant (2026-06-22T09:47:03.413Z)

All findings are fully verified against the live repos. Now let me apply the surgical edits.

****Finding 1 (medium): Status DRAFT / Last updated ? flip to IN PROGRESS, bump date.**** P0-P4 are merged (#15/#22), so per the lifecycle this is at least IN PROGRESS (P5-P9 remain owner-gated).

12. user (2026-06-22T09:47:05.578Z)

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13. assistant (2026-06-22T09:47:09.368Z)

****Finding 3 (high): "indexed in docs/README.md" claimed three times, but NOT actually indexed.****
The audit confirms `docs/README.md` has no entry. Per the rule, I must keep status markers honest without inventing a completion. I cannot add the entry to `docs/README.md` (that's a different file). So I correct the doc's claims to reflect it is not yet indexed.

Line 5 (tracking anchors):

14. user (2026-06-22T09:47:11.181Z)

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15. assistant (2026-06-22T09:47:15.369Z)

Now line 71, the \$7 P0 row claims "indexed in `docs/README.md`" as ?. The deliverable was "This tracked doc in `khelsutra/docs/`, ****indexed in `docs/README.md`****". The doc exists (#15) but is not indexed. I'll mark it ? (In progress) and note the indexing is still pending.

16. user (2026-06-22T09:47:17.933Z)

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17. assistant (2026-06-22T09:47:23.139Z)

Now line 91, the DoD bullet "\$7 tracker reflects real shipped state; doc indexed in `docs/README.md`." Since it's not indexed, leave the checkbox unchecked (it already is ` - [ ] `) but the claim is fine as a DoD requirement (it's an unchecked aspiration, which is honest). The audit's concern is the ***claims of completion***, not the DoD requirement. Let me re-read line 91 ? it's ` - [ ] ` (unchecked), so it correctly states it's still to do. No edit needed there.

****Finding 2 (high): P5 marked Todo though annotator-UI screenshots committed (#25).**** Split: the annotator-UI screenshots are done (#25); footage screenshots with consented people remain Todo.

18. user (2026-06-22T09:47:26.054Z)

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19. assistant (2026-06-22T09:47:28.720Z)

****Finding 6 (medium):** §7 "(this PR)" placeholders ? cite #22 (and #23 for org-link).****** P1-P4 shipped in #22.

20. user (2026-06-22T09:47:34.730Z)

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21. assistant (2026-06-22T09:47:37.230Z)

****Finding 5 (low):** "lands with PR #13" future-tense ? the doc exists now.****** Two spots: line 6 (Relation) and line 84 (Nice-to-knows).

22. user (2026-06-22T09:47:40.375Z)

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23. user (2026-06-22T09:47:44.872Z)

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24. assistant (2026-06-22T09:47:48.018Z)

****Finding 4 (low):** SPA "one static screen" framing predates the wizard (#88).****** Reword line 53 to reflect App.tsx now renders a first-run SetupWizard. The "no router" sub-claim is still true (no react-router dep).

25. user (2026-06-22T09:47:50.158Z)

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26. assistant (2026-06-22T09:47:57.192Z)

Now the low-severity ****line-citation drifts**** (facts correct, citations stale). These are flagged stale findings with suggestedFix to update citations. Let me apply them.

****gate_verdict harness.py:98 ? :142**** ? appears at line 12 (TL;DR) and line 31 (§3).

27. user (2026-06-22T09:47:58.628Z)

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28. user (2026-06-22T09:48:00.494Z)

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29. assistant (2026-06-22T09:48:03.495Z)

****cmd_train __main__.py:231 ? :332**** (lines 31, 49, 95), and the ****join citation `\_\_main\_\_.py:271-280` ? :383-384**** (lines 49, 95). Let me handle line 31 first.

30. user (2026-06-22T09:48:05.135Z)

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31. assistant (2026-06-22T09:48:08.724Z)

Now line 49: the join citation `\_\_main\_\_.py:271-280`. On master the stem-join is at `:383-384`. Let me update it.

32. user (2026-06-22T09:48:10.855Z)

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33. assistant (2026-06-22T09:48:17.043Z)

****no_paid_llm_training_labels :192 ? :236, label_source :178 ? :222.**** These appear at line 25 (D6: `harness.py:178,189-192`), line 31 (§3: `harness.py:178`? no ? line 33 actually), line 33 (Guardrails), line 60 (threat model), and line 95 (References).

Let me check the exact strings. Line 25 (D6): `harness.py:178,189-192`. Let me verify the attestation block range on master.

34. user (2026-06-22T09:48:18.288Z)

```
=== attestation block context ===
1             "honest generalization = the LOVO eval below"},
2         "training_lineage": {
3             "corpus_manifest": manifest_path,
4             "split_protocol": "leave-one-video-out by match",
5             "label_source": "human (rally-annotator golden CSVs)",
6             "videos": [{"name": s["name"], "labels": s["labels"],
7                         "gt_hash": gt_hash(load_gt_intervals(s["labels"],
strict=False)),
8                         "fps": s["fps"]}] for s in specs],
9         },
10        "eval": {
11            "protocol": "LOVO, F1@tIoU=0.5 segment-level + per-frame AUC",
12            "per_video": eval_res["per_video"],
13            "mean_f1": eval_res["mean_f1"],
14            "floor": floor,
15        },
16        "attestation": {
17            "deps_licenses": "numpy/scipy (BSD) + repo-internal code only (Gen-0)",
18            "wasb_checkpoint_provenance": "C3-UNRESOLVED (ship-blocking; see
COMMERCIALIZATION C3)",
19            "no_paid_llm_training_labels": True,
20            "consent_ledger": "pending-M0.1 (owner-attested: owned footage, all-adult
sessions)",
21        },
22        "gate": verdict,
23    }
```

35. assistant (2026-06-22T09:48:21.667Z)

So `label\_source` is at :222, and the `attestation` block spans :233-237 with `no\_paid\_llm\_training\_labels` at :236. The doc's D6 cites `harness.py:178,189-192` ? `:178` was label_source (now :222), `:189-192` was the attestation block (now :233-237). Let me update.

Line 25 (D6):

36. user (2026-06-22T09:48:23.739Z)

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37. assistant (2026-06-22T09:48:26.712Z)

Line 33 (Guardrails): `no\_paid\_llm\_training\_labels:True`, `harness.py:192` and `harness.py:178`.

38. user (2026-06-22T09:48:28.482Z)

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39. assistant (2026-06-22T09:48:31.419Z)

Line 60 (threat model): `attested in the bundle (harness.py:192)`.

40. user (2026-06-22T09:48:33.830Z)

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41. assistant (2026-06-22T09:48:37.695Z)

****server.js alias :49-51 ? :99**** (line 52) and the `site/src/index.js:84` Worker reference (still correct). Also the OG image `site/og.png` ? `site/public/og.png` (line 52). Both are in the same Reuse-map bullet (line 52).

42. user (2026-06-22T09:48:41.048Z)

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43. assistant (2026-06-22T09:48:46.157Z)

Now the ****References section (line 95)**** still cites old engine line numbers: `cmd\_train :231`, `gate\_verdict :98`, `label\_source :178`, `attestation :189-192`. Let me update these for consistency. Let me re-read the current References line.

44. user (2026-06-22T09:48:46.870Z)

```
94      ## 10. References
95      **Annotator [verified]** `rally-annotator/README.md`, `vlc/rally_annotator.lua`
(HEADER :128; now_seconds :201-207; marks :608-628; save_all :310-335; resolve_out_path
:253-267), `docs/CSV_FORMAT.md`, `docs/ENDING_REASONS.md`, `LICENSE` (MIT),
`.github/workflows/ci.yml`.
96      **Engine [verified]** `backend/worker/__main__.py` (cmd_train :231;
_write_intervals_csv :62-74; join :271-280; harness shell-out :321-328),
`backend/pipeline/detectors/stub_runner.py` (synth :124-149, :131-133, :139),
`training/gen0/harness.py` (gate_verdict :98; label_source :178; attestation :189-192),
`backend/eval/gt_loader.py` (:14-90), `docs/SETUP_NEW_MACHINE.md`,
`docs/DECOUPLED_COMPUTE/02-COMPUTE-MAP-AND-CONTRACT.md`, `docs/DECOUPLED_COMPUTE/04-BUILD-AND-
TEST-PLAN.md`, `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`, `CLAUDE.md`.
97      **KhelSutra [verified]** `site/public/index.html`, `site/public/capture.html`,
`site/server.js`, `site/src/index.js`, `web/src/App.tsx`, `web/package.json`, `docs/README.md`,
`docs/PER_RALLY_SELECTION.md`.
```

45. assistant (2026-06-22T09:48:53.540Z)

The Engine references line cites several drifted anchors. Let me verify the remaining ones I haven't confirmed: `\_write\_intervals\_csv` (:72 on master, doc says :62-74), `harness shell-out` (:424 on master, doc says :321-328), `stub\_runner synth :124-149/:131-133/:139`, `gt\_loader :14-90`.

I confirmed earlier: `\_write\_intervals\_csv` is at :72 (master), `harness shell-out` command at :424, `synth\_trajectory\_from\_labels` at :124, DictReader :137, start_time :139, gt_loader load_gt_intervals :44 / _START_NAMES :30. Let me get the precise ranges for the shell-out and _write_intervals_csv to update accurately.

46. user (2026-06-22T09:48:56.692Z)

```
=== _write_intervals_csv range ===
1 def _write_intervals_csv(segments: List[dict], path: str) -> None:
2     """Decimal-second `[start,end)` rallies + shot count / ending reason ? the join-
friendly
3     artifact (same schema as the rally-annotator labels)."""
4     with open(path, "w", newline="") as f:
5         w = csv.writer(f)
6         w.writerow(["start", "end", "shot_count", "ending_reason"])
7         for s in segments:
8             w.writerow([
9                 f'{float(s.get("start_time", 0.0)):.3f}',
10                f'{float(s.get("end_time", 0.0)):.3f}',
11                s.get("shot_count", ""),
12                s.get("ending_reason", s.get("shot_count_confidence", "")),
13            ])
14
15
16 def _write_report_json(video_id: str, segments: List[dict], status: str, path: str) ->
None:
17     with open(path, "w") as f:
18         json.dump({
19             "video_id": video_id,
=== harness shell-out cmd range ===
1     cmd = [sys.executable, "-m", "training.gen0.harness",
2             "--manifest", manifest_path, "--out", out_dir, "--version", args.version]
3     if args.floor:
4         floor_local = (storage.fetch(args.floor, os.path.join(scratch, "floor.json"),
config=storage_cfg)
5                         if "://" in args.floor else args.floor)
6         cmd += ["--floor", floor_local]
7     print("[worker] training:", " ".join(cmd))
8     rc = subprocess.run(cmd).returncode
9     if rc != 0:
=== gt_loader end ===
1: def load_gt_intervals(csv_path: str, *, strict: bool = True) -> List[Interval]:
=== stub_runner synth end ===
1: def synth_trajectory_from_labels(labels_csv: str, out_csv: str, *, fps: float = 30.0,
6:     cover the gaps ? so the synthetic trajectory *encodes* the rally labels.
```

47. assistant (2026-06-22T09:49:01.990Z)

Now I have the precise anchors. Let me also confirm `\_write\_intervals\_csv` header now uses `start,end,shot\_count,ending\_reason` ? yes, confirmed at the line above (matches doc line 48 claim). The `\_write\_intervals\_csv` is at :72-85, harness shell-out at :424-432.

The `\_write\_intervals\_csv` :62-74` doc citation at line 49 (`__main__.py:62-74`) ? let me check: that's referenced in line 49 of the doc. Actually line 49 references `_main__.py:62-74` for the worker's inference CSV. Let me re-read line 49.

The doc line 49 says: `The worker's own inference CSV uses start,end,shot\_count,ending\_reason (\_\_main\_\_.py:62-74)`. On master `\_write\_intervals\_csv` is :72-85. Let me update that and the References line.

48. user (2026-06-22T09:49:03.879Z)

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49. assistant (2026-06-22T09:49:10.124Z)

Now the References line (96). Updating the drifted engine anchors: cmd_train :231?:332, _write_intervals_csv :62-74?:72-85, join :271-280?:383-384, harness shell-out :321-328?:424-432, gate_verdict :98?:142, label_source :178?:222, attestation :189-192?:233-237. The stub_runner :124-149/:131-133/:139 ? synth function spans roughly :124-149 still (DictReader :137, append :139), I'll leave those as the audit didn't flag them and :139 is confirmed. gt_loader :14-90 ? load_gt_intervals at :44, _START_NAMES at :30; the doc earlier (line 47) cites `gt\_loader.py:14-90` which the audit marked correct, so I leave that range.

50. user (2026-06-22T09:49:14.017Z)

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51. assistant (2026-06-22T09:49:21.712Z)

Now let me handle two remaining citation drifts I noted in the body that the audit didn't explicitly enumerate but are part of the same "citations drifted" findings:

- Line 31 §3 also cites `stub\_runner.py:124-149,131-133` ? the audit confirmed `:124` and `:139`; `:131-133` is internal. I'll leave the §3 prose range as the audit didn't flag it specifically and `:124-149` is still the function span.
- Line 49 cites `stub\_runner.py:139` for the DictReader path ? confirmed correct on master (:139).

Let me also check line 12 still references `stub\_runner.py:124-149,131-133` and line 95 changelog. Those weren't flagged. Now let me update the Changelog to record this freshness pass, and verify there are no other `(this PR)` or stale references left.

52. user (2026-06-22T09:49:24.512Z)

```
=== remaining (this PR) ===  
=== remaining :98 / :231 / :178 / :192 / :49-51 / PR #13 ===  
=== site/og.png (bare) ===
```

53. assistant (2026-06-22T09:49:26.454Z)

All stale tokens cleared. Now let me add a Changelog entry to record this freshness pass honestly.

54. user (2026-06-22T09:49:33.469Z)

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55. assistant (2026-06-22T09:49:37.762Z)

Now let me verify the DoD bullet at line 91 once more, and double-check the §7 P0 row change didn't break the "indexed in docs/README.md" claim in DoD. Let me re-read the relevant sections to confirm consistency.

56. user (2026-06-22T09:49:38.576Z)

```
86  
87     ## 9. Definition of done  
88     - [ ] `own-model.html` live with Annotate/Train/Setup sections, honesty fences, verbatim  
repo text, GitHub link, real (non-fabricated) screenshots.  
89     - [ ] `own-model` alias works on both the Node host and the Cloudflare Worker; cross-  
linked from nav/section/footer.  
90     - [ ] Every claim tagged `[verified]`/[design]`/[researched]`; no accuracy-superiority  
claim; no "Train now" button; VLC-3.0.x-only + GPU/native-WASB-gated + owner-gated stated  
plainly.  
91     - [ ] Schema-bridge + MD5/stem-join prerequisite documented; the two CSV schemas kept
```

distinct.

92 - [] \$7 tracker reflects real shipped state; doc indexed in `docs/README.md`.

57. assistant (2026-06-22T09:49:43.114Z)

The DoD bullets are all unchecked `- [ ]`, which is honest ? line 92 correctly states "doc indexed in `docs/README.md`" as a still-unmet DoD requirement. No edit needed there; the prior claim-of-completion problems were at lines 5 and 71, both now corrected.

All findings for `docs/ANNOTATE\_AND\_TRAIN.md` are addressed. The one code change that would be needed (adding the actual index entry to `docs/README.md`) is out of scope for a docs-only edit of my assigned file. Let me record that under `codeChangesNeeded` and submit the report.

58. user (2026-06-22T09:50:15.105Z)

Structured output provided successfully

59. assistant (2026-06-22T09:50:20.559Z)

Completed the freshness sweep for `docs/ANNOTATE\_AND\_TRAIN.md`. All six stale findings were re-verified against the live engine and khelsutra `origin/master` checkouts before editing, then applied as surgical edits preserving the doc's voice and structure. One follow-up (adding the actual index entry to `docs/README.md`) is recorded under `codeChangesNeeded` since it falls outside my assigned file.